

Formatting String

- C-Style String Formatting
- `format ( )` Method
- f-String / Formatted String Literals

C-Style String Formatting

% operator/ Modulo Operator/ Interpolation Operator – This operator is used for formatting strings. It interprets the left argument much like a `sprintf( )` style format string to be applied to the right argument, and returns the string resulting from this formatting operation.

Syntax:- `print("format placeholder" %(data))`

Format placeholder = `%[flags][width][.precision]type`

% - marks the start of the specifier

Flags – It affect the result of some conversion type

Width – Minimum field width

Precision – Given as . Followed by the precision

Type – Conversion type

C-Style String Formatting

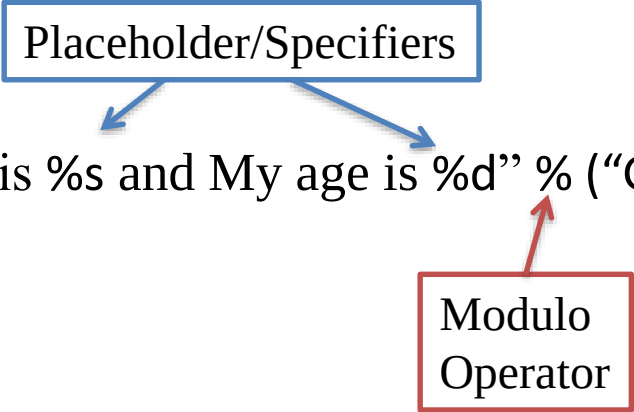
Syntax:- `print("format placeholder" %(data))`

Format placeholder = `%[flags][width][.precision]type`

data – It can be literal, variable, expression etc.

Ex:-

Placeholder/Specifiers



```
print("My name is %s and My age is %d" % ("GeekyShows", 62))
```

Modulo
Operator

Maintain Order in above statement first string then integer

C-Style String Formatting

Syntax:- `print("format placeholder" % {'key':value})`

Format placeholder = `%(mapping_key)[flags][width][.precision]type`

% - marks the start of the specifier

Mapping_key – Consisting of parenthesized sequence of characters

Flags – It affect the result of some conversion type

Width – Minimum field width

Precision – Given as . Followed by the precision

Type – Conversion type

C-Style String Formatting

Example:-

```
print("My name is %(nm)s and My age is %(ag)d" % {'nm':"GeekyShows", 'ag':62})
```

```
print("My name is %(nm)s and My age is %(ag)d" % {'ag':62, 'nm':"GeekyShows"})
```

Do not need to maintain Order in above statement.

C-Style String Formatting

Conversion Type	Meaning
d	Signed integer decimal.
I	Signed integer Decimal
o	Signed octal value.
x	Signed hexadecimal (lowercase).
X	Signed hexadecimal (uppercase).
e	Floating point exponential format (lowercase).
E	Floating point exponential format (uppercase).
f	Floating point decimal format.
F	Floating point decimal format.

C-Style String Formatting

Conversion Type	Meaning
g	Floating Point format. Uses lower case exponential format if exponent is less than -4 or not less than precision, decimal format otherwise
G	Floating Point format. Uses lower case exponential format if exponent is less than -4 or not less than precision, decimal format otherwise
c	Single character (accepts integer or single character string)
r	String (converts any python object using repr())
s	String (converts any Python object using str()).
A	String (Converts any python object using ascii())
%	No argument is converted, result in a % character in the result

C-Style String Formatting

Flag	Meaning
#	Used with o, x or X specifiers the value is preceded with 0, 0o, 0O, 0x or 0X respectively.
0	The conversion will be zero padded for numeric values.
-	The converted value is left adjusted (overrides the '0' conversion if both are given).
	(a space) A blank should be left before a positive number (or empty string) produced by a signed conversion.
+	A sign character ('+' or '-') will precede the conversion (overrides a “space” flag).

C-Style String Formatting

<code>print("%d" % 432)</code>	432
<code>print("%d %d" % (432, 345))</code>	432 345
<code>print("%f" % 432.123)</code>	432.123000
<code>print("%f" % 432.123456)</code>	432.123456
<code>print("%f" % 432.12345651)</code>	432.123457
<code>print("%f" % 432.12345641)</code>	432.123456
<code>print("%s" % "GeekyShows")</code>	GeekyShows
<code>print("%s %s" % ("Hello", "GeekyShows"))</code>	Hello GeekyShows
<code>print("%d %s" % (432, "GeekyShows"))</code>	432 GeekyShows
<code>#print("%s %d" % (432, "GeekyShows"))</code>	TypeError
<code>print("%(nm)s %(ag)d" % {'ag':432, 'nm':"GeekyShows"})</code>	GeekyShows 432

C-Style String Formatting

`print("%d" % 432)`

432

`print("%+d" % 432)`

+432

`print("%8d" % 432)`

432

					4	3	2
--	--	--	--	--	---	---	---

`print("%08d" % 432)`

00000432

0	0	0	0	0	4	3	2
---	---	---	---	---	---	---	---

C-Style String Formatting

`print("%f" % 432.123)`

432.123000

`print("%.3f" % 432.123)`

4	3	2	.	1	2	3
---	---	---	---	---	---	---

`print("%.2f" % 432.123)`

4	3	2	.	1	2
---	---	---	---	---	---

`print("%.2f" % 432.128)`

4	3	2	.	1	3
---	---	---	---	---	---

C-Style String Formatting

```
print("%9.2f" % 432.128)
```

			4	3	2	.	1	3
--	--	--	---	---	---	---	---	---

```
print("%09.2f" % 432.123)
```

0	0	0	4	3	2	.	1	2
---	---	---	---	---	---	---	---	---

```
print("%9.2f" % 4388453232.124)
```